

Weekly Research Report — Week 1

MITACS Research Internship

Project: Developing Machine Learning GraphRAG Augmenting Techniques for Clinical Case Description and Patient Journey Follow-Ups

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Summary of Work Done This Week

This was the first week of the MITACS research engagement. The primary focus was on foundational literature review — conducting a detailed, structured analysis of two of the six core reference papers assigned for this project. In parallel, the project GitHub repository was initialized with a folder structure for tracking weekly progress, paper notes, and experimental notebooks.

Paper 1 — Sesen, Au Yeung & Asgari (2025)

Full citation: Sesen, M.B., Au Yeung, J., & Asgari, E. (2025). *Development and validation of Retrieval Augmented Generation (RAG) and GraphRAG for complex clinical cases*. medRxiv. <https://doi.org/10.1101/2025.11.25.25341010>

Problem Statement

The paper addresses the challenge of applying complex clinical guidelines (NICE CKD guidelines) to individual patient data using LLMs. Clinicians face cognitive overload when interpreting guidelines for unique patient contexts, and baseline LLMs are prone to hallucinations and lack local or institutional guideline knowledge.

What They Built

Three clinical decision support (CDS) pipelines were compared:

- (1) **Baseline LLM** — GPT-4o with no augmentation.
- (2) **LLM + Vector RAG** — GPT-4o with NICE CKD guidelines chunked and vector-indexed using `llama_index`.
- (3) **LLM + GraphRAG** — GPT-4o with the same guidelines converted into a knowledge graph using Microsoft's open-source GraphRAG package.

Dataset

- 70 synthetic CKD patients, each with 17 structured clinical features (age, GFR, blood pressure, potassium, bicarbonate, haemoglobin, etc.)
- 9 clinician-curated questions per patient (630 total QA pairs)
- Question types ranged from binary determinate tasks (referral decisions) to open-ended management plans and numerical formula tasks (KFRE risk score)

- Gold-standard answers validated by a consultant nephrologist

Knowledge Graph Specifications

- Built from NICE CKD guideline text using Microsoft GraphRAG
- **208 nodes, 280 edges** — sparse graph reflecting NICE’s modular structure
- Three major clusters: NICE, CKD, and Chronic Kidney Disease as central hubs
- Edge types represented clinical actions: “used,” “treatment,” “monitoring,” “provides”
- Visualized using Gephi with modularity-based community detection
- Used **local search at community level 2** for patient-specific queries

Key Results

Table 1: Performance comparison across three CDS pipelines (Sesen et al., 2025)

Metric	Baseline LLM	Vector RAG	GraphRAG
Clinical correctness (clinician)	70.8%	76.6%	75.6%
Patient-specificity score	2.35	3.77	6.59
Clarity score (1–10)	9.01	8.53	7.80
Best on structured lookups	—		
Best on open-ended management	—	—	
Best on out-of-corpus questions		—	—

Critical Limitations and Research Gaps for My Project

- Gap 1: No ontology normalization.** “CKD” and “Chronic Kidney Disease” were treated as separate nodes, fragmenting the graph. The authors explicitly recommend adding UMLS/SNOMED normalization. My project directly addresses this.
- Gap 2: Single guideline indexed.** Only NICE CKD guidelines were used. Key drug classes (SGLT-2 inhibitors, finerenone) were missing from the corpus. My project will index multiple sources.
- Gap 3: Synthetic patients only.** No real EHR data was used. My project targets real clinical case descriptions structured using the SOAP framework.
- Gap 4: No SOAP structure.** Patient data was tabular (17 flat features). My project structures patient data as typed SOAP nodes in Neo4j, which is richer and more clinically realistic.
- Gap 5: Clarity vs. specificity trade-off unresolved.** GraphRAG produced verbose outputs due to long guideline excerpts. A post-processing summarization step was suggested but never implemented — a concrete improvement I plan to build.

Technical Elements to Borrow

- Microsoft GraphRAG package (<https://github.com/microsoft/graphrag>)
- llama_index Python framework for RAG pipeline
- LLM-as-Judge evaluation methodology alongside clinician scoring
- Separate prompts per question per patient (not batched) — avoids token limit issues
- Community level = 2 for local search as a starting parameter
- Gephi for knowledge graph visualization

Paper 2 — Chen et al. (2026), MED-COPILOT

Full citation: Chen, S., Patil, N., Pan, H., Hwang, A.H., Du, Y., Liu, R., & Zhao, J. (2026). *MED-COPILOT: A Medical Assistant Powered by GraphRAG and Similar Patient Case Retrieval*. arXiv:2603.00460. <https://arxiv.org/abs/2603.00460>

Live demo: https://huggingface.co/spaces/Cryo3978/Med_GraphRAG

Problem Statement

Clinical decisions require two types of evidence that prior systems do not jointly provide: (1) structured guideline knowledge, and (2) outcomes from clinically similar past patients. Existing RAG systems focus on guidelines only and ignore patient-level analogical reasoning.

The Two-Pillar System

Pillar 1 — GraphRAG on guidelines:

- 118 WHO + 525 NICE guidelines = 40 million tokens
- Knowledge graph with entities: diseases, symptoms, medications, procedures
- Relations: indication, contraindication, monitoring, escalation
- Normalized to **SNOMED-CT, ICD-10, ATC**
- Embedded using **BioClinicalBERT**, stored in **LanceDB**
- Community-level summarization for efficient retrieval

Pillar 2 — Similar Patient Retrieval (SPR):

- 36,000 patient cases total
- 18,000 from MIMIC-IV (real ICU data, de-identified)
- 18,000 from Synthea (synthetic patient generator)
- All converted to **SOAP format** using a rule-guided conversion agent
- Hybrid retrieval: keyword matching (diagnoses, comorbidities) + semantic embedding

Table 2: Ablation results isolating retrieval contributions (Chen et al., 2026, GPT-4.1-mini backbone)

Setting	ROUGE-L	BERTScore-F1	MedQA Acc	MMLU-clin Acc
Baseline (no retrieval)	0.144	0.825	0.846	0.868
+ RAG only	0.221	0.831	0.885	0.887
+ SPR only	0.159	0.828	0.851	0.914
+ GraphRAG only	0.258	0.840	0.870	0.952
+ Both (full)	0.291	0.851	0.912	0.971

Key Results

Key finding: GraphRAG alone outperforms SPR alone, but combining both yields the strongest results by a significant margin — the two sources are *complementary*, not redundant.

Graph Construction Pipeline (Technical Detail)

1. Segment guideline into TextUnits (one recommendation per unit, not arbitrary windows)
2. Extract entities and relations using medical NER and LLM prompting
3. Normalize entities to SNOMED-CT, ICD-10, ATC vocabularies
4. Encode constraint qualifiers (age limits, escalation triggers) explicitly
5. Build community summaries for efficient high-level retrieval
6. Embed with BioClinicalBERT → LanceDB vector store

Comparison with Paper 1

Critical Limitations and Research Gaps

Gap 1: Heterogeneous data distributions. MIMIC-IV and Synthea have different documentation styles. A normalization quality check would strengthen results.

Gap 2: Graph loses contextual nuance. Converting narrative guidelines to structured relations drops some meaning.

Gap 3: No uncertainty quantification. The system does not report confidence levels alongside recommendations.

Gap 4: No gold standard for patient similarity. There is no benchmark to formally evaluate whether retrieved similar patients are clinically appropriate.

Table 3: Key differences between Sesen et al. (2025) and Chen et al. (2026)

Aspect	Sesen et al.	Chen et al.
Guidelines indexed	1 (NICE CKD only)	643 (WHO + NICE)
Patient data	70 synthetic, tabular	36,000 SOAP-structured
Patient structure	Flat 17 features	Full SOAP format
Ontology	None	SNOMED-CT, ICD-10, ATC
Similar patient retrieval	No	Yes (hybrid)
Embedding model	Not specified	BioClinicalBERT
Vector store	llama_index default	LanceDB
Public demo	No	Yes (HuggingFace)

Technical Elements to Borrow

- SOAP conversion pipeline (rule-guided agent for EHR → SOAP)
- Dual retrieval architecture (GraphRAG + SPR combined)
- BioClinicalBERT for clinical text embeddings
- MIMIC-IV dataset (requires credentialing — apply now, takes 2–3 weeks)
- HuggingFace Spaces as the public demo delivery format
- Evaluation on ROUGE-L, BERTScore-F1, MedQA, MMLU-clinical benchmarks

Synthesis: My Project’s Unique Contributions

Neither paper alone matches the full scope of this MITACS project. The table below maps gaps from both papers to my planned contributions:

Table 4: Research gap to contribution mapping

Gap in existing work	Source	My contribution
No UMLS ontology normalization	Sesen et al.	UMLS entity linking in Neo4j KG
Tabular patient features only	Sesen et al.	Typed SOAP nodes in KG
Single guideline indexed	Sesen et al.	Multi-source guideline indexing
No uncertainty quantification	Chen et al.	Planned hallucination rate metric
No follow-up order generation	Both papers	Novel P-section generation task
Clarity vs. specificity gap	Sesen et al.	Post-processing summarization layer

GitHub Repository

mitacs-graphrag-clinical/

```
README.md
reading_log.md          <- Paper 1 and 2 notes committed
weekly_reports/
  week_01.md            <- Markdown version of this report
notebooks/              <- Experimental IPYNBs (from Week 2)
papers/                 <- Reference links (no PDFs)
```

Plan for Next Week (Week 2)

1. Read and document **Paper 3** — Henry & Gong (2026), KOSMOS ontology scaffolding (directly relevant to UMLS integration)
2. Read and document **Paper 4** — Baptista & Garcia (2025), GraphRAG on EHR
3. Set up local development environment: Neo4j Desktop, ChromaDB, LangChain/LlamaIndex, Microsoft GraphRAG package
4. Apply for **MIMIC-III/IV data access** (credentialing takes 2–3 weeks — apply immediately)
5. Push `week01_environment_test.ipynb` to GitHub

Questions / Blockers for Supervisor

1. Do we have institutional API access (OpenAI / Azure OpenAI) or a budget allocation for LLM API calls during experiments?
 2. Should I apply for MIMIC-III/IV access now, or will we use a different clinical dataset?
 3. Is there a preferred target IEEE venue or conference deadline I should plan the paper submission around?
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